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How Cookie Consent Interfaces Changed

A Traceable Longitudinal Audit With a Controlled Pilot and Six Dated Cases

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Abstract

Cookie-consent interfaces are often assessed through a single banner or screenshot. That perspective cannot establish whether refusal becomes easier, whether a user can understand the choice, or whether the same interface changes over time. This paper presents a traceable longitudinal audit framework with four components: path availability, path effort, transparency, and choice effectiveness. The framework is applied in two evidence lanes. A controlled five-site pilot tests whether the capture, versioning, and scoring workflow can preserve comparable observations; because it has only one validated interval, all five local directional labels remain insufficient evidence. A separate, purposively selected case series reconstructs six dated company trajectories from primary regulatory decisions, regulator follow-ups, and one company announcement. Five cases improve at least one audited component and one later regresses. In all three first-layer cases, refusal moves from an acceptance-favoring path to one-click or equivalently simple rejection. Orange and SHEIN document technical remediation of refusal or withdrawal, while Vanity Fair shows that a later functional failure can follow an earlier closed proceeding. The findings do not estimate web-wide prevalence or establish a single causal effect of regulation. They support a narrower conclusion: concrete interactions that regulators can specify and verify are the clearest site of documented improvement, while information quality and technical respect for refusal can remain incomplete or reversible.

1. Introduction

Cookie banners make a consequential choice look deceptively simple. A visitor may see an Accept button, a Reject button, a link to settings, or no refusal route at all. The visible first layer is not the whole decision: a refusal can require several additional actions, the consequences can be unclear, and a site can continue reading or writing consent-dependent cookies after rejection. An audit that treats the first screenshot as the complete object of study can therefore miss the pathway and behavioral parts of the decision.

This problem also has a temporal dimension. A changed DOM, screenshot, or button label is evidence that an interface changed, not evidence that user choice improved. To support a directional claim, the two states must be dated and comparable, the same decision components must be assessed at both points, and the result must be distinguished from any proposed explanation for why the change occurred. This paper develops that distinction as a practical research method.

The work is positioned alongside, rather than as a replacement for, broad historical measurement. Dimova et al. examined archived banners from 11,364 websites in 30 countries and found that the share offering both Accept and Reject rose from 2.94 percent in 2018 to 30.66 percent in 2024 [1]. Their archive-based result supplies an important population-level trend, with an explicit lower-bound caveat. The contribution here is narrower: a traceable component rubric applied to company-level, dated change points, with direction and explanatory strength recorded separately.

The project asks two linked questions:

1. How can a computational audit and scoring system quantify layered consent interfaces in terms of unbiased choice across the full consent pathway?

2. How can firms' privacy interfaces be captured and versioned to document change systematically over time?

The first question supplies the ruler; the second supplies the timeline. The paper does not report a randomized experiment, issue legal compliance verdicts, or infer an internet-wide improvement rate from the selected cases.

2. Method

2.1 Component rubric

Each consent state is assessed with four user-facing components.

- Path availability: whether a user can reach acceptance, rejection, and customization.

- Path effort: the number and character of actions required for each route, including whether refusal has parity with acceptance.

- Transparency: whether purposes and consequences are presented clearly.
- Choice effectiveness: whether refusal or withdrawal governs the relevant technical behavior, including cookie reads and writes.

The rubric is deliberately not one scalar privacy score. A page can improve on one component and remain weak on another. For example, a first-layer reject button can remove effort asymmetry while the surrounding information remains unclear; conversely, an apparently balanced interface cannot demonstrate that a refusal changes backend behavior.

For a dated pair, an improvement requires at least one audited component to improve with no assessed component regressing. A regression requires the reverse; mixed is reserved for opposing component changes; insufficient evidence applies when states are not comparable or the evidence cannot support the comparison. A raw technical difference alone is not a directional result.

2.2 Two evidence lanes

The controlled pilot tested the capture, versioning, and comparison workflow across five sites over one validated May 29--June 5 interval. It produced a preserved, reviewable capture package, but one interval cannot establish a defensible local trajectory. All five labels therefore remain insufficient_evidence. This is a method result: it demonstrates why capture context, scorer version, and evidence review must be preserved before an interface difference becomes a substantive claim.

The retrospective case series answers a different question. A case was included only when it named a company and consent surface, described two dated states in one jurisdiction, supplied primary evidence for both states, supported coding of at least one rubric component, and allowed observed direction to be separated from the reason for change. The resulting six cases are purposively selected source-complete trajectories, not a representative sample of firms or websites.

2.3 Evidence strength

Direction and explanation are graded separately. Direction is strong when a regulator records both states or verifies remediation. It is moderate when the comparison uses a company announcement for one state or a broader regulator-recorded compliance transition. Explanation is recorded as direct company attribution, regulator-verified order response, change during an investigation or proceeding, or unknown. This guardrail prevents timing alone from being converted into a universal causal claim.

3. Results

3.1 Controlled pilot

The checked-in local package contains 42 audit reports and 20 longitudinal summaries. The five matched-site reviews remain directionally insufficient. This conservative result is substantive: the study does not recode missing time points, same-day repeats, capture failures, or changed scorer behavior as evidence that a consent interface became better or worse.

3.2 Six dated company trajectories

Company and surface	Earlier dated state	Later dated state	Component result	Direction / explanation evidence
Google Search and YouTube	Acceptance took one action; rejection took at least five	Equal first-screen Reject all and Accept all actions	Availability and effort improved	Moderate split-source direction; direct company attribution [3,4]
Facebook	Immediate acceptance had no refusal	Equivalent refusal was deployed and CNIL	Availability, effort, and verified operation	Strong direction; regulator-verified order

Company and surface	Earlier dated state	Later dated state	Component result	Direction / explanation evidence
	mechanism of equivalent simplicity	closed the injunction	improved	response [5,6]
TikTok	Acceptance took one action; rejection took at least three	A first-layer Tout refuser button was added	Availability and effort improved; purpose information remained deficient	Strong direction; investigation-period change [7]
Orange	Cookies continued to be read after withdrawal	First-party cookies were removed and new third-party requests stopped	Withdrawal effectiveness improved	Strong direction; regulator-verified order response [8]
Vanity Fair France	An earlier compliance proceeding closed	Later checks found pre-choice cookies and continued behavior after rejection or withdrawal	Refusal effectiveness regressed; transparency not assessed	Moderate state-transition direction; reason unknown [9]
SHEIN	Cookies were observed before choice and after refusal or withdrawal	The authority recorded refusal and withdrawal remediation during proceedings	Technical refusal and withdrawal effectiveness improved	Strong direction; proceedings-period change [10]

Table source: retrospective case table and registered primary-source registry, both dated July 29, 2026.

Across the six selected cases, five improve at least one audited component and one regresses. The three first-layer cases all reduce acceptance-favoring rejection effort: Google introduces equal first-screen actions, Facebook deploys equivalent refusal, and TikTok adds a first-layer rejection button. Those fractions summarize this case series only and are not prevalence estimates.

The component distinction matters within the positive cases. Google provides the clearest direct attribution: the company stated that its revised Europe cookie choices followed updated regulatory guidance and specific CNIL direction [4]. Facebook and Orange have a different but still strong pattern: a regulator ordered a correction and later verified the changed mechanism [6,8]. TikTok and SHEIN document a change during regulatory activity, but the available records do not prove that the investigation or proceeding was the only cause [7,10].

The result is not a simple story of visual improvement. TikTok's first-layer refusal reduced effort while the regulatory decision still found cookie-purpose information insufficient [7]. Orange and SHEIN address the technical effect of refusal or withdrawal, which cannot be inferred from button labels alone. Vanity Fair supplies the counterexample: later checks found cookies before choice and continued operations after rejection or withdrawal even though an earlier proceeding had been closed [9]. The later regression is documented; its cause is not. Because the earlier record lacks a comparable transparency baseline, transparency direction is not assessed.

4. Discussion

The repeated pattern is that concrete user interactions improve most clearly when an authority can specify and verify them. Equal refusal, one-click rejection, and a cessation of specified cookie operations after withdrawal are all observable remedies. This is compatible with, but does not prove, a wider influence of regulatory scrutiny. The evidence is strongest for Google's own attribution and the Facebook and Orange order-and-follow-up records; it is weaker for changes that merely occurred during a regulatory process.

The findings also explain why a single visual audit is not enough. A button that looks balanced does not establish whether the user understood the choice or whether the choice changed data behavior. Conversely, a later technical failure may not be visible in the banner at all. A defensible audit should thus keep availability, effort, transparency, and choice effectiveness visible as separate components and should repeat the assessment rather than treating an earlier closure or a polished banner as permanent certification.

The methodology offers a practical bridge between controlled web capture and historical evidence. Local capture supplies reproducible context and can reveal why a short interval is not yet adequate for a trajectory. Dated regulator and company records can recover longer company histories where preserved local browser states are unavailable. The two lanes should not be merged into one sample: they have different provenance and answer different questions.

5. Limitations

The local pilot has only one validated interval, so it does not support a local directional conclusion. The historical cases were selected because their public records were source-complete; their 5/6 and 1/6 fractions cannot estimate the prevalence of improvement across the web. The cases are concentrated in a French regulatory setting, and their later states use mixed primary-source forms: regulator decisions, regulator follow-ups, and one company announcement. Finally, the study does not establish that regulation alone caused each observed change, and it does not make a legal determination beyond the dated observations recorded by the cited sources.

6. Conclusion

The project treats RQ1 as a repeatable component measure and RQ2 as the timeline needed to interpret change. The controlled pilot validates a cautious capture-and-comparison workflow while remaining directionally insufficient. The six-case series shows that the same rubric can recover dated, source-linked evolution: five cases improve at least one component and one functionally regresses. The appropriate conclusion is not that cookie-consent interfaces universally improved. It is that concrete, verifiable choice mechanisms are the clearest site of documented improvement, while transparency and technical respect for refusal remain component-specific and reversible.

References

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